

Digital Integrated Circuit Design

HW-1

Problem:

An IC design house is very interested in a simple ALU. They want to purchase this ALU, but they ask us to add some more features. Please design a more powerful ALU based on the simple ALU in HW-1 with the following specifications:

- 1.1 The demand of the 8-bit ALU as Fig. 1 and a 4-bit instruction set as shown in Table 1.

Table 1 4-bit instruction set

Instruction[3:0]	Operation	Notes
0000	2's complement addition	$X=A+B$
0001	2's complement subtraction	$X=A-B$
0010	NOT	$X=\sim B$
0011	AND	$X=A\&B$
0100	OR	$X=A\mid B$
0101	XOR	$X=A\wedge B$
others	No operation	--

- 1.2 For instructions 0000 and 0001, they do not permit using one adder and one subtractor separately. Instead, please use only **one** 8-bit adder and **one** 8-bit **incrementer** (ex. $D=C+8'b1$).
- 1.3 Besides, the **symbol “~”** occurs in your RTL **only once**.
- 1.4 Please write a test bench based on HW-1 to test your RTL code in an exhaustive way. Of course, we will use our own test bench to judge your ALU.

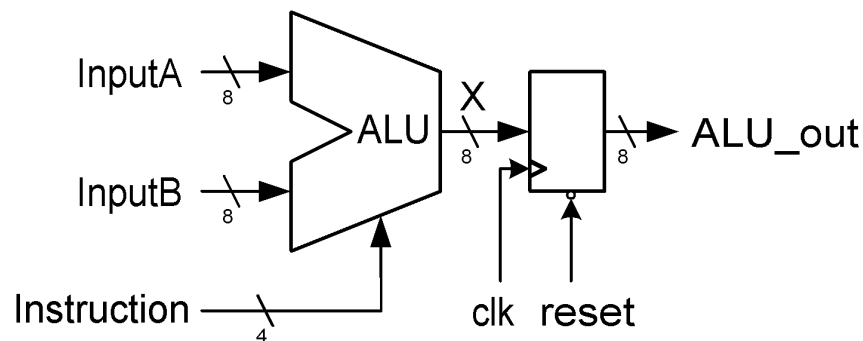


Fig. 1 8-bit ALU

Submission

- Please submit the following files to E-learning.
 - B123456_HW1_rtl.v**: Your top-module design, where **B123456** is your **student ID**. If you have more than two RTL codes, for instance, please name as B123456_DECODER_rtl.v.
 - B123456_HW1_rtb.v
- Please deliver the simulation results and the circuit diagrams (including logic-gate circuit) of your design (.docx) in class.
 - For the simulation results, please attach the related photos (.jpg or pdf) in .docx.
 - For the circuit diagrams (including logic-gate circuit), please sketch it using VISIO (.vsdx) or pen and then please scan it (.jpg or .pdf). After that, please paste the circuit diagram in .docx.
 - Please enlarge your photos and circuit diagrams to make them clear.